

CONJOINT ANALYSIS

The term “conjoint analysis” refers to a set of techniques for indirectly estimating preferences in general and consumer price sensitivity in particular. These techniques are indirect because the consumer is never asked: “How much are you willing to pay for this product?” Instead, the consumer is shown two or more product descriptions and asked to indicate which descriptions he or she prefers. The consumer’s price sensitivity is then inferred from his or her stated preferences for different product descriptions.

In the 1970s, one popular way to obtain conjoint estimates involved handing respondents a stack of 40 or more cards, each containing a unique product description, and asking the respondents to rank the cards from the most preferred to the least preferred. Marketers then analyzed the respondent rankings to determine why respondents preferred some concepts over others. We will use this methodology to introduce the basics of conjoint analysis.

Step One

Select a product or service that you are thinking about buying in the near future. For example, you might select a coffeemaker, a computer, a printer, a mountain bike, a DVD player, cell phone, car, or vacation. (You might also think about a nontraditional product such as the kind of summer internship that you would like.) Identify five attributes of this product or service. For three of these attributes, identify 3 attribute levels, and for the remaining 2 attributes, identify two attribute levels.

For example, suppose we choose a car. Five car attributes are mileage, body style, horsepower, engine size, and price. The following table contains three attribute levels for the first three attributes, and two attribute levels for the remaining attributes:

This exercise was prepared by Mark E. Parry, Professor of Business Administration, Darden Graduate School of Business Administration, University of Virginia. The approach of this exercise was adopted from Fred S. Zufryden’s article, “Course Evaluation and Design Optimization: A Conjoint Analysis-Based Application,” *Interfaces*, Linticum; Vol. 13, No. 2 (April 1983): 87-95. Copyright © 2002 by the University of Virginia Darden School Foundation, Charlottesville, VA. All rights reserved. To order copies, send an e-mail to sales@ardenpublishing.com. No part of this publication may be reproduced, stored in a retrieval system, used in a spreadsheet, or transmitted in any form or by any means—electronic, mechanical, photocopying, recording, or otherwise—without the permission of the Darden School Foundation.

Table 1					
Attribute Levels	Attribute 1: Mileage	Attribute 2: Body Style	Attribute 3: Horsepower	Attribute 4: Engine Size	Attribute 5: Price
1	8/12	4-Door Sedan	172	4 cylinder	\$18,700
2	15/22	2-Door Sedan	185	6 cylinder	\$27,500
3	25/37	Convertible	220	—	—

(There are two numbers in the mileage column, because manufacturers provide two mileage estimates, one for city driving and the other for highway driving.)

Step Two

Use the following chart to create five product concepts by combining attribute levels according to the following chart:

Table 2					
Concept	Attribute 1: Mileage	Attribute 2: Body Style	Attribute 3: Horsepower	Attribute 4: Engine Size	Attribute 5: Price
1	1	1	1	1	1
2	1	2	3	1	2
3	2	1	3	2	1
4	2	3	2	1	1
5	3	3	1	2	2

Thus your first concept should consist of level 1 of each of the five attributes that you selected.

Using the attributes and attribute levels in Table 1, we obtain the following five concepts:

Table 3					
Concept	Attribute 1: Mileage	Attribute 2: Body Style	Attribute 3: Horsepower	Attribute 4: Engine Size	Attribute 5: Price
1	8/12	4-Door Sedan	172	4 cylinder	\$18,700
2	8/12	2-Door Sedan	220	4 cylinder	\$27,500
3	15/22	4-Door Sedan	220	6 cylinder	\$18,700
4	15/22	Convertible	185	4 cylinder	\$18,700
5	25/37	Convertible	172	6 cylinder	\$27,500

Use the tables in **Exhibit 1** to create 5 product concept cards, 1 for each concept. Rank these 5 concepts from 1 to 5. Let 5 denote your most preferred concept and 1 denote your least preferred concept. Record your rankings on the holdout concept column in **Exhibit 3**.

Step Three

Now create 18 additional product concepts by combining attribute levels according to the following chart:

Table 4					
Concept	Attribute One	Attribute Two	Attribute Three	Attribute Four	Attribute Five
A	1	1	1	2	2
B	1	2	2	2	1
C	1	3	3	1	2
D	2	1	2	1	1
E	2	2	3	2	2
F	2	3	1	2	2
G	3	1	3	2	1
H	3	2	1	1	2
I	3	3	2	2	2
J	1	1	3	1	2
K	1	2	1	2	1
L	1	3	2	2	2
M	2	1	1	2	2
N	2	2	2	1	2
O	2	3	3	2	1
P	3	1	2	2	2
Q	3	2	3	2	2
R	3	3	1	1	1

Thus concept A should consist of level 1 of attributes 1, 2, and 3, and level 2 of attributes 4 and 5.

Using the attributes and attribute levels in Table 1, we obtain the following eighteen concepts:

Table 5					
Concept	Attribute 1: Mileage	Attribute 2: Body Style	Attribute 3: Horsepower	Attribute 4: Engine Size	Attribute 5: Price
A	8/12	4-Door Sedan	172	6 cylinder	\$27,500
B	8/12	2-Door Sedan	185	6 cylinder	\$18,700
C	8/12	Convertible	220	4 cylinder	\$27,500
D	15/22	4-Door Sedan	185	4 cylinder	\$18,700
E	15/22	2-Door Sedan	220	6 cylinder	\$27,500
F	15/22	Convertible	172	6 cylinder	\$27,500
G	25/37	4-Door Sedan	220	6 cylinder	\$18,700
H	25/37	2-Door Sedan	172	4 cylinder	\$27,500
I	25/37	Convertible	185	6 cylinder	\$27,500
J	8/12	4-Door Sedan	220	4 cylinder	\$27,500
K	8/12	2-Door Sedan	172	6 cylinder	\$18,700
L	8/12	Convertible	185	6 cylinder	\$27,500
M	15/22	4-Door Sedan	172	6 cylinder	\$27,500
N	15/22	2-Door Sedan	185	4 cylinder	\$27,500
O	15/22	Convertible	220	6 cylinder	\$18,700
P	25/37	4-Door Sedan	185	6 cylinder	\$27,500
Q	25/37	2-Door Sedan	220	6 cylinder	\$27,500
R	25/37	Convertible	172	4 cylinder	\$18,700

Use the tables in **Exhibit 2** to create 18 product concept cards, 1 for each of the concepts created from Table 4. Cut these cards out of the Exhibit, so that you have one piece of paper for each concept. Sort these 18 concepts into 3 piles: a most-preferred pile, a least-preferred pile, and an “in-between” pile (concepts in the in-between pile are not as good as concepts in the most-preferred pile, but are better than concepts in the least-preferred pile.) Finally, sort the concepts within each pile from the most-preferred concept to the least-preferred concept. On the basis of your sorting, rank the 18 concepts and record these rankings in the estimation concept column in **Exhibit 3**. Let 18 denote your most-preferred concept and 1 denote your least-preferred concept.

Step Four

We will use regression analysis to determine the importance of different attributes and different attribute levels. The dependent variable in the regression will be the rankings that you recorded in **Exhibit 3**. The independent variables will be a series of dummy variables that are defined as follows:

A_{12} = 1 if the concept contains level 2 of attribute 1, 0 otherwise;

A_{13} = 1 if the concept contains level 3 of attribute 1, 0 otherwise;

A_{22} = 1 if the concept contains level 2 of attribute 2, 0 otherwise;

A_{23} = 1 if the concept contains level 3 of attribute 2, 0 otherwise;

A_{32} = 1 if the concept contains level 2 of attribute 3, 0 otherwise;

A_{33} = 1 if the concept contains level 3 of attribute 3, 0 otherwise;

A_{42} = 1 if the concept contains level 2 of attribute 4, 0 otherwise; and

A_{52} = 1 if the concept contains level 2 of attribute 5, 0 otherwise.

Notice the notation: dummy variable A_{ij} is the dummy variable for attribute level j of attribute i .

Using these definitions, we can create the following independent variables:

Concept	A12	A13	A22	A23	A32	A33	A42	A52
A	0	0	0	0	0	0	1	1
B	0	0	1	0	1	0	1	0
C	0	0	0	1	0	1	0	1
D	1	0	0	0	1	0	0	0
E	1	0	1	0	0	1	1	1
F	1	0	0	1	0	0	1	1
G	0	1	0	0	0	1	1	0
H	0	1	1	0	0	0	0	1
I	0	1	0	1	1	0	1	1
J	0	0	0	0	0	1	0	1
K	0	0	1	0	0	0	1	0
L	0	0	0	1	1	0	1	1
M	1	0	0	0	0	0	1	1
N	1	0	1	0	1	0	0	1
O	1	0	0	1	0	1	1	0
P	0	1	0	0	1	0	1	1
Q	0	1	1	0	0	1	1	1
R	0	1	0	1	0	0	0	0

We can use these dummy variables as independent variables in a regression. For example, suppose that a respondent offered the following rankings of the holdout concepts in Table 3 and the estimation concepts in Table 5.

Holdout Concepts		Estimation Concepts	
Concept	Ranking	Concept	Ranking
1	3	A	3
2	1	B	14
3	5	C	2
4	4	D	16
5	2	E	7
		F	5
		G	18
		H	10
		I	9
		J	4
		K	13
		L	1
		M	8
		N	6
		O	15
		P	12
		Q	11
		R	17

We put the estimation rankings together with these dummy variables in an Excel spreadsheet. The resulting spreadsheet looks like this:

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	A	B	C	D	E	F	G	H	I	J	K
1	Concept	Ranking	A12	A13	A22	A23	A32	A33	A42	A52	
2	A	3	0	0	0	0	0	0	1	1	
3	B	14	0	0	1	0	1	0	1	0	
4	C	2	0	0	0	1	0	1	0	1	
5	D	16	1	0	0	0	1	0	0	0	
6	E	7	1	0	1	0	0	1	1	1	
7	F	5	1	0	0	1	0	0	1	1	
8	G	18	0	1	0	0	0	1	1	0	
9	H	10	0	1	1	0	0	0	0	1	
10	I	9	0	1	0	1	1	0	1	1	
11	J	4	0	0	0	0	0	1	0	1	
12	K	13	0	0	1	0	0	0	1	0	
13	L	1	0	0	0	1	1	0	1	1	
14	M	8	1	0	0	0	0	0	1	1	
15	N	6	1	0	1	0	1	0	0	1	
16	O	15	1	0	0	1	0	1	1	0	
17	P	12	0	1	0	0	1	0	1	1	
18	Q	11	0	1	1	0	0	1	1	1	
19	R	17	0	1	0	1	0	0	0	0	

Sheet1 Sheet2 Sheet3

Ready

Now regress the rankings in Column B on the dummy variables. The following table contains the regression results using the variables in the spreadsheet on the previous page:

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	L	M	N	O	P	Q	R	S	T
10	Regression Statistics			ANOVA					
11	Multiple R	0.99			df	SS	MS	F	Significance F
12	R Square	0.98		Regression	8	431.18	53.90	50.23	0.00
13	Adjusted R Square	0.96		Residual	8	8.58	1.07		
14	Standard Error	1.04		Total	16	439.76			
15	Observations	17.00							
16									
17	Variable	Level		Coefficients	Standard Error	t Stat	P-value		
18			Intercept	12.75	0.90	14.21	0.00		
19	Mileage	15/22	A12	3.08	0.64	4.81	0.00		
20	Mileage	25/37	A13	6.42	0.64	10.01	0.00		
21	Body Style	2-Door Sedan	A22	-0.25	0.64	-0.39	0.71		
22	Body Style	Convertible	A23	-2.25	0.64	-3.51	0.01		
23	Horsepower	185	A32	0.08	0.64	0.13	0.90		
24	Horsepower	220	A33	-0.08	0.64	-0.13	0.90		
25	Engine Size	6-Cylinder	A42	0.62	0.53	1.18	0.27		
26	Price	\$27,500	A52	-8.88	0.53	-16.72	0.00		
27									

Sheet1 Sheet2 Sheet3

Ready

Notice that the intercept—12.75—indicates the ranking of a product concept where every attribute level equals one. Let's call this the *base concept*. In our example, the base concept is a 4-door sedan with a 4 cylinder engine that generates 172 horsepower, costs \$18,700, and gets 8 miles per gallon with city driving and 12 miles per gallon with highway driving.

The regression coefficients tell us how the ranking of the base concept will change if we change one attribute level. For example, suppose that, instead of getting 8 miles per gallon in the city, the car gets 15 miles per gallon. The coefficient for variable A12 indicates that the predicted ranking will rise by 3.08, from 12.75 to 15.83. If the car gets 25 miles per gallon in the city, the attribute A13 indicates that the predicted ranking will rise from 12.75 to 19.17.

Step Five

Use the regression coefficients to predict the rankings of your holdout concepts. To do this, start with the base concept ranking and adjust that ranking using the regression coefficients. For example, the first holdout concept in Table 2 has every attribute level equal to 1. Thus the predicted ranking is simply the ranking of the base concept, 12.75. The second holdout concept is described as follows:

- + Attribute 1, Level 1
- + Attribute 2, Level 2
- + Attribute 3, Level 3
- + Attribute 4, Level 1
- + Attribute 5, Level 2.

Thus the predicted ranking will be:

$$\begin{aligned} & \text{Intercept} + \text{Coefficient of A22} + \text{Coefficient of A33} + \text{Coefficient of A52} \\ & = 12.75 + 0 - 0.25 - 0.08 - 8.88 = 3.54 \end{aligned}$$

Doing this by hand is tedious. To save time, we can recast the holdout concept descriptions using the dummy variables defined above. We obtain:

Table 6

	Attribute 1: Mileage		Attribute 2: Body Style		Attribute 3: Horsepower		Attribute 4: Engine Size	Attribute 5: Price
Concept	A12	A13	A22	A23	A32	A33	A42	A52
1	0	0	0	0	0	0	0	0
2	0	0	1	0	0	1	0	1
3	1	0	0	0	0	1	1	0
4	1	0	0	1	1	0	0	0
5	0	1	0	1	0	0	1	1

We use these dummy variables along with the dummy variable regression coefficients to predict the rankings for the holdout concepts.

Returning to our earlier example, we can set up an Excel spreadsheet to look like this:

	A	B	C	D	E	F	G	H	I	J	K
	Holdout Concept	Attribute 1: Mileage	Attribute 2: Body Style	Attribute 3: Horsepower	Attribute 4: Engine Size	Attribute 5: Price				Predicted Ranking	Holdout Ranking
1	1	0	0	0	0	0	0	0	0	12.75	3
2	2	0	0	1	0	0	1	0	1	3.54	1
3	3	1	0	0	0	0	1	1	0	16.38	5
4	4	1	0	0	1	1	0	0	0	13.58	4
5	5	0	1	0	1	0	0	1	1	8.67	2
7	Dummy Variable	A12	A13	A22	A23	A32	A33	A42	A52		
8	Regression Coefficient	3.08	6.42	-0.25	-2.25	0.08	-0.08	0.62	-8.88		
9	Intercept	12.75									
10											
11											
12											
13											
14											
15											
16											
17											

Column J contains the predicted rankings, which are obtained by multiplying the dummy variables by the appropriate regression coefficients, summing the results, and adding the intercept coefficient in cell B9. For example, the formula in cell J2 is:

$$=B2*\$B\$8+C2*\$C\$8+D2*\$D\$8+E2*\$E\$8+F2*\$F\$8+G2*\$G\$8+H2*\$H\$8+I2*\$I\$8+\$B\$9$$

Notice that the predicted rankings range from 3.54 to 16.38. The magnitude of these numbers reflects the fact that the regression coefficients were obtained from a regression in which the dependent variable ranged from 1 to 18. For this reason, we cannot directly compare the predicted rankings in column J with the rankings supplied by our hypothetical respondent (column K). What we can do is compare the *rank order* implied by the numbers in column J with the rank order recorded in column K. When we do this, we find that the rank order of the “predicted rankings” exactly matches the order of the rankings supplied by our hypothetical

respondent. This suggests that the regression coefficients do a reasonable job of summarizing the preferences of our hypothetical respondent.

Analysis Assignment

1. Regress your rankings of the 18 product concepts on the 8 dummy variables defined above. Interpret your results.
2. Use the holdout concepts to evaluate the usefulness of the conjoint utilities that you have estimated.

Exhibit 1

CONJOINT ANALYSIS

Holdout Concepts

Holdout Concept 1		
Attribute	Attribute Level	Attribute Level Name
1	1	
2	1	
3	1	
4	1	
5	1	

Holdout Concept 2		
Attribute	Attribute Level	Attribute Level Name
1	1	
2	2	
3	3	
4	1	
5	2	

Holdout Concept 3		
Attribute	Attribute Level	Attribute Level Name
1	2	
2	1	
3	3	
4	2	
5	1	

Exhibit 1 (continued)

Holdout Concepts

Holdout Concept 4		
Attribute	Attribute Level	Attribute Level Name
1	2	
2	3	
3	2	
4	1	
5	1	

Holdout Concept 5		
Attribute	Attribute Level	Attribute Level Name
1	3	
2	3	
3	1	
4	2	
5	2	

Exhibit 2

CONJOINT ANALYSIS

Product Concepts for Estimating Product Utilities

Concept A		
Attribute	Attribute Level	Attribute and Attribute Level Name
1	1	
2	1	
3	1	
4	2	
5	2	

Concept B		
Attribute	Attribute Level	Attribute and Attribute Level Name
1	1	
2	2	
3	2	
4	2	
5	1	

Concept C		
Attribute	Attribute Level	Attribute and Attribute Level Name
1	1	
2	3	
3	3	
4	1	
5	2	

Exhibit 2 (continued)

Product Concepts for Estimating Product Utilities

Concept D		
Attribute	Attribute Level	Attribute and Attribute Level Name
1	2	
2	1	
3	2	
4	1	
5	1	

Concept E		
Attribute	Attribute Level	Attribute and Attribute Level Name
1	2	
2	2	
3	3	
4	2	
5	2	

Concept F		
Attribute	Attribute Level	Attribute and Attribute Level Name
1	2	
2	3	
3	1	
4	2	
5	2	

Exhibit 2 (continued)

Product Concepts for Estimating Product Utilities

Concept G		
Attribute	Attribute Level	Attribute and Attribute Level Name
1	3	
2	1	
3	3	
4	2	
5	1	

Concept H		
Attribute	Attribute Level	Attribute and Attribute Level Name
1	3	
2	2	
3	1	
4	1	
5	2	

Concept I		
Attribute	Attribute Level	Attribute and Attribute Level Name
1	3	
2	3	
3	2	
4	2	
5	2	

Exhibit 2 (continued)

Product Concepts for Estimating Product Utilities

Concept J		
Attribute	Attribute Level	Attribute and Attribute Level Name
1	1	
2	1	
3	3	
4	1	
5	2	

Concept K		
Attribute	Attribute Level	Attribute and Attribute Level Name
1	1	
2	2	
3	1	
4	2	
5	1	

Concept L		
Attribute	Attribute Level	Attribute and Attribute Level Name
1	1	
2	3	
3	2	
4	2	
5	2	

Exhibit 2 (continued)

Product Concepts for Estimating Product Utilities

Concept M		
Attribute	Attribute Level	Attribute and Attribute Level Name
1	2	
2	1	
3	1	
4	2	
5	2	

Concept N		
Attribute	Attribute Level	Attribute and Attribute Level Name
1	2	
2	2	
3	2	
4	1	
5	2	

Concept O		
Attribute	Attribute Level	Attribute and Attribute Level Name
1	2	
2	3	
3	3	
4	2	
5	1	

Exhibit 2 (continued)

Product Concepts for Estimating Product Utilities

Concept P		
Attribute	Attribute Level	Attribute and Attribute Level Name
1	3	
2	1	
3	2	
4	2	
5	2	

Concept Q		
Attribute	Attribute Level	Attribute and Attribute Level Name
1	3	
2	2	
3	3	
4	2	
5	2	

Concept R		
Attribute	Attribute Level	Attribute and Attribute Level Name
1	3	
2	3	
3	1	
4	1	
5	1	

Exhibit 3

CONJOINT ANALYSIS

Data Collection Form

Holdout Concepts		Estimation Concepts	
Concept	Ranking	Concept	Ranking
1		A	
2		B	
3		C	
4		D	
5		E	
		F	
		G	
		H	
		I	
		J	
		K	
		L	
		M	
		N	
		O	
		P	
		Q	
		R	